Lecture 10: Chapter 7 Review-
Strategy of Integration

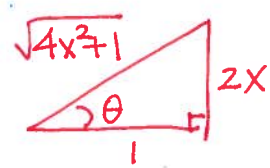
ex. $\displaystyle\int \frac{1}{x\sqrt{4x^2+1}}\,dx$ [Trig-sub.] $2x = \tan\theta$

$(\ln|\frac{\sqrt{4x^2+1}}{2x} - \frac{1}{2x}| + c)$ $2dx = \sec^2\theta\, d\theta$

$$= \int \frac{1}{\frac{1}{2}\tan\theta \cdot \sec\theta} \cdot \frac{1}{2}\sec^2\theta\, d\theta$$

$$= \int \frac{\sec\theta}{\tan\theta}\, d\theta$$

$$= \int \frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta}\, d\theta$$



$$= \int \csc\theta\, d\theta = \ln|\csc\theta - \cot\theta| + C$$

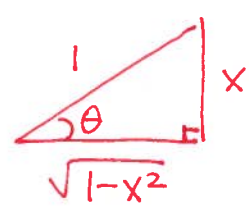
$$= \ln\left|\frac{\sqrt{4x^2+1}}{2x} - \frac{1}{2x}\right| + C$$

ex. $\displaystyle\int \frac{dx}{(1-x^2)^{3/2}}$ [trig-sub.] $x = \sin\theta$

$(\frac{x}{\sqrt{1-x^2}} + c)$ $dx = \cos\theta\, d\theta$

$$= \int \frac{\cos\theta\, d\theta}{\cos^3\theta}$$

$$= \int \sec^2\theta\, d\theta$$



$$= \tan\theta + C$$

$$= \frac{x}{\sqrt{1-x^2}} + C$$

Strategy for integration:

1. Simplify the integrand if possible;
making sure rational expression is **proper**.

2. Look for obvious **substitution**.

3. If integrand is a product involving a polynomial, trig, exponential, logarithmic functions , try **by parts**.

4. If $f(x)$ is a product of powers of $\sin x$ and $\cos x$, $\tan x$ and $\sec x$, $\cot x$ and $\csc x$, then we use the rules for **trig. integration**.

5. If $f(x)$ is a rational function that's not obvious how to integrate, try **partial fraction**.

6. If $\sqrt{\pm a^2x^2 \pm b^2}$ occurs, try **trig.sub**;

7. If $\sqrt[n]{g(x)}$ occurs, rationalize it by using a **u-sub**.

8. Sometimes, you need to manipulate the integrand, use several methods. Just try and try.

ex. $\displaystyle\int \frac{\sin^3 x}{\cos x}\,dx$ [u-sub] $u = \cos x$

$(\frac{\cos^2 x}{2} - \ln|\cos x| + c)$ $du = -\sin x\,dx$

$$= \int \frac{\sin^2 x \cdot \sin x}{\cos x}\,dx$$

$$= \int \frac{(1-u^2)\cdot(-du)}{u}$$

$$= \int \frac{u^2-1}{u}\,du = \int u - \frac{1}{u}\,du = \frac{1}{2}u^2 - \ln|u| + C$$

$$= \frac{1}{2}\cos^2 x - \ln|\cos x| + C$$

ex. $\displaystyle\int \frac{\sec^3 x}{\tan^2 x}\,dx$ [$\sec^2 x = 1 + \tan^2 x$]

$(-\csc x + \ln|\sec x + \tan x| + c)$

$$= \int \frac{\sec^2 x \cdot \sec x}{\tan^2 x}\,dx = \int \frac{(1+\tan^2 x)\sec x}{\tan^2 x}\,dx$$

$$= \int \frac{\sec x}{\tan^2 x} + \frac{\cancel{\tan^2 x}\sec x}{\cancel{\tan^2 x}}\,dx$$

$$= -\frac{1}{\sin x} + \ln|\sec x + \tan x| + C$$

$$\int \frac{\sec x}{\tan^2 x}\,dx = \int \frac{1}{\cos x}\cdot\frac{\cos^2 x}{\sin^2 x}\,dx$$

$$= \int \frac{\cos x}{\sin^2 x}\,dx \qquad u = \sin x$$

$$= \int \frac{1}{u^2}\,du$$

$$= -\frac{1}{u}$$

ex. $\displaystyle\int \frac{\arctan x}{x^2}\,dx$ [IBP]

$\left(-\frac{\arctan x}{x} + \ln|x| - \frac{1}{2}\ln(1+x^2) + c\right)$

$u = \tan^{-1}x \qquad dv = \frac{1}{x^2}dx$

$du = \frac{1}{1+x^2} \qquad v = -\frac{1}{x}$

$$= -\frac{1}{x}\tan^{-1}x + \int \frac{1}{x(1+x^2)}\,dx$$

PFD: $\frac{1}{x(1+x^2)} = \frac{A}{x} + \frac{Bx+C}{1+x^2}$

cover-up: $x=0$, $A=1$

$$\frac{1}{x(1+x^2)} = \frac{1}{x} + \frac{Bx+C}{1+x^2}$$

$$1 = (1+x^2) + (Bx+C)x$$

$$1 = (1+B)x^2 + Cx + 1$$

$B=-1$, $C=0$

$$= -\frac{1}{x}\tan^{-1}x + \int \frac{1}{x} - \frac{x}{1+x^2}\,dx$$

$$= -\frac{1}{x}\tan^{-1}x + \ln|x| - \frac{1}{2}\ln(1+x^2) + C$$

or combine two u-subs into one : $u = x^{3/2}$

ex. $\displaystyle\int \frac{\sqrt{x}}{1+x^3}\,dx$ $\boxed{\text{[rationalizing sub]}}$

$(\frac{2}{3}\arctan(x^{\frac{3}{2}}) + c)$

$u = \sqrt{x}$

$u^2 = x$

$2u\,du = dx$

$$= \int \frac{u}{1+u^6}\,2u\,du$$

$$= 2\int \frac{u^2}{1+u^6}\,du$$

$$= 2\int \frac{u^2}{1+(u^3)^2}\,du$$

u-sub: $t = u^3$

$dt = 3u^2\,du$

$$= \frac{2}{3}\int \frac{dt}{1+t^2}$$

$$= \frac{2}{3}\tan^{-1}(t) + C$$

$$= \frac{2}{3}\tan^{-1}(u^3) + C$$

$$= \frac{2}{3}\tan^{-1}(\sqrt{x})^3 + C$$

Lecture 10 – NYTI

1. $\displaystyle\int \frac{dx}{1+2e^x-e^{-x}}$ $\boxed{[\ u=e^x\ ]}$

$((1/3)ln|(2e^x-1)|-(1/3)ln|e^x+1|+c)$ $du=e^x dx$, $\frac{du}{e^x}=dx$, $\frac{du}{u}=dx$

$$= \int \frac{1}{1+2u-\frac{1}{u}} \cdot \frac{du}{u}$$

$$= \int \frac{1}{2u^2+u-1}\, du$$

PFD: $\frac{1}{(2u-1)(u+1)} = \frac{A}{2u-1} + \frac{B}{u+1}$

$A = \frac{1}{\frac{1}{2}+1} = \frac{2}{3}$

$B = \frac{1}{-2-1} = -\frac{1}{3}$

$$= \int \frac{2/3}{2u-1} - \frac{1/3}{u+1}\, du$$

$$= \frac{1}{3}\ln|2u-1| - \frac{1}{3}\ln|u+1| + C = \frac{1}{3}\ln|2e^x-1| - \frac{1}{3}\ln|e^x+1| + C$$

2. $\displaystyle\int \frac{e^{2x}}{1+e^x}\, dx$ $\boxed{[\ u=e^x\ ,\ du=e^x dx\ ]}$, $\frac{du}{u}=dx$

$(e^x - \ln(1+e^x)+c)$

$$= \int \frac{u^2}{1+u} \cdot \frac{du}{u}$$

long division: $u+1 \overline{\big)\, u}$ quotient 1, remainder -1

$$= \int \frac{u}{u+1}\, du$$

$$= \int 1 - \frac{1}{u+1}\, du$$

$$= u - \ln|u+1| + C$$

$$= e^x - \ln(e^x+1) + C$$

3. $\displaystyle\int \frac{\ln(x+1)}{x^2}\,dx$ [IBP]

$(-\frac{\ln|x+1|}{x} + \ln|x| - \ln|x+1| + c)$

$u = \ln(x+1)$ $\quad + \quad dv = \frac{1}{x^2}dx$

$du = \frac{1}{x+1}dx \quad - \quad v = -\frac{1}{x}$

$= -\frac{1}{x}\ln(x+1) + \int \frac{1}{x(x+1)}dx$

PFD: $\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$

$A = 1, \; B = -1$

$= -\frac{1}{x}\ln(x+1) + \int \frac{1}{x} - \frac{1}{x+1}\,dx$

$= -\frac{1}{x}\ln(x+1) + \ln|x| - \ln(x+1) + C$

4. $\displaystyle\int \frac{4^x + 10^x}{2^x}\,dx$ [Simplify]

$(\frac{2^x}{\ln 2} + \frac{5^x}{\ln 5} + c)$

$= \int \frac{4^x}{2^x} + \frac{10^x}{2^x}\,dx$

$= \int 2^x + 5^x \; dx$

$= \frac{2^x}{\ln 2} + \frac{5^x}{\ln 5} + C$

5. $\displaystyle\int \frac{dx}{\sqrt{x}(2+\sqrt{x})^4}$ $\boxed{[\quad u=2+\sqrt{x} \quad]}$ $du=\frac{1}{2\sqrt{x}}dx$

$(-\frac{2}{3}(2+\sqrt{x})^{-3}+c)$

$2\sqrt{x}\,du=dx$

$$= \int \frac{2\cancel{\sqrt{x}}\,du}{\cancel{\sqrt{x}}\;u^4}$$

$$= \int \frac{2}{u^4}\,du$$

$$= \frac{2}{-3}u^{-3}+C = -\frac{2}{3}\frac{1}{(2+\sqrt{x})^3}+C$$

6. $\displaystyle\int \sqrt{1+e^x}\,dx$ ① $\boxed{[\quad u=e^x,\ du=e^x dx \quad]}$, $\frac{du}{u}=dx$

$(2\sqrt{1+e^x}+\ln|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}|+c)$

$$= \int \sqrt{1+u}\cdot\frac{du}{u}$$

② $v=\sqrt{1+u}$

$v^2=1+u$, $2v\,dv=du$

$$= \int \frac{v\cdot 2v\,dv}{v^2-1}$$

$$= 2\int \frac{v^2}{v^2-1}\,dv$$

③ long division $v^2-1 \overline{\big)\, v^2}$ quotient 1, v^2-1, remainder 1

$$= 2\int 1+\frac{1}{v^2-1}\,dv$$

④ PFD $\frac{1}{(v+1)(v-1)}=\frac{A}{v+1}+\frac{B}{v-1}$

$A=-\frac{1}{2}$, $B=\frac{1}{2}$

$$= 2\int 1+\frac{1/2}{v-1}-\frac{1/2}{v+1}\,dv$$

$$= 2v+\ln|v-1|-\ln|v+1|+C = 2\sqrt{1+e^x}+\ln|\sqrt{1+e^x}-1|-\ln|\sqrt{1+e^x}+1|+C$$

$u = \sin x, \quad du = \cos x \, dx$

7. $\displaystyle\int \frac{\cos^5 x}{\sqrt{\sin x}} dx$ $2\sqrt{\sin x} - \frac{4\sin^{5/2} x}{5} + \frac{2\sin^{9/2} x}{9} + c$

$$= \int \frac{\cos^4 x}{\sqrt{\sin x}} \cos x \, dx$$

$$= \int \frac{(1-u^2)^2}{\sqrt{u}} du$$

$$= \int \frac{u^4 - 2u^2 + 1}{\sqrt{u}} du$$

$$= \int u^{7/2} - 2u^{3/2} + u^{-1/2} du = 2u^{1}$$

8. $\displaystyle\int \frac{\sin^3(\sqrt{x})}{\sqrt{x}} dx$ $\frac{2\cos^3\sqrt{x}}{3} - 2\cos\sqrt{x} + c$

$u = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}} dx$

$2\sqrt{x}\, du = dx$

$$= \int \frac{\sin^3(u)}{\sqrt{x}} \cdot 2\sqrt{x}\, du$$

$$= 2\int \sin^3 u \, du$$

$$= 2\int \sin^2 u \cdot \sin u \, du$$

$v = \cos u$

$dv = -\sin u \, du$

$$= 2\int (1-v^2)\cdot(-dv)$$

$$= 2\int v^2 - 1 \, d$$

$$= 2\left(\tfrac{1}{3}v^3 - v\right) + C = \tfrac{2}{3}\cos^3(\sqrt{x}) - 2\cos(\sqrt{x}) + C$$

9. $\displaystyle\int \frac{\cos x + \sin x}{\sin(2x)}\,dx$ $\left(\frac{1}{2}(\ln|\csc x - \cot x| + \ln|\sec x + \tan x|) + c\right)$

Simplify!

$$= \int \frac{\cos x + \sin x}{2\sin x \cos x}\,dx$$

$$= \int \frac{\cancel{\cos x}}{2\sin x \cancel{\cos x}} + \frac{\cancel{\sin x}}{2\cancel{\sin x}\cos x}\,dx$$

$$= \int \frac{1}{2}\csc x + \frac{1}{2}\sec x\,dx$$

$$= \frac{1}{2}\ln|\csc x - \cot x| + \frac{1}{2}\ln|\sec x + \tan x| + C$$

10. $\displaystyle\int \frac{1 - \tan^2 x}{\sec^2 x}\,dx$ $\left(\frac{1}{2}\sin(2x) + c\right)$

Simplify!

$$= \int \frac{1}{\sec^2 x} - \frac{\tan^2 x}{\sec^2 x}\,dx$$

$$= \int \cos^2 x - \sin^2 x\,dx$$

$$= \int \cos(2x)\,dx$$

$$= \frac{1}{2}\sin(2x) + C$$

11. $\displaystyle\int \frac{dx}{\cos x - 1} \cdot \frac{\cos x + 1}{(\csc x + \cot x + c)\ \cos x + 1}$

$$= \int \frac{\cos x + 1}{\cos^2 x - 1}\, dx$$

$$= \int \frac{\cos x + 1}{-\sin^2 x}\, dx$$

$$= \int \frac{\cos x}{-\sin^2 x} - \frac{1}{\sin^2 x}\, dx$$

$$= \int -\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} - \frac{1}{\sin^2 x}\, dx$$

$$= \int -\cot x \csc x - \csc^2 x\, dx$$

$$= +\csc x + \cot x + C$$